

Composites of Biopolymers and ZnO NPs for Controlled Release of Zinc in Agricultural Soils and Timed Delivery for Maize

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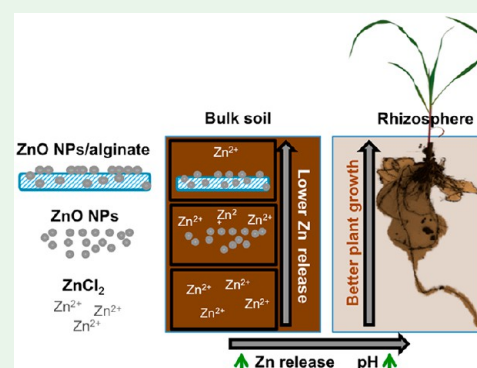
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ABSTRACT: Zinc (Zn) micronutrient deficiency is a widespread problem affecting crops worldwide. Unlike conventional ionic fertilizers (Zn as salt or chelated forms), Zn-based engineered nanomaterials (ENMs) have the potential to release Zn in a controlled manner, reducing Zn losses through leaching upon application to soil. In this work, composites made of biopolymers (microcrystalline cellulose, chitosan, and alginate) and ZnO nanoparticles [NPs; 4–65% (w/w) Zn] were prepared. Their potential for Zn controlled release was tested in four agricultural soils of distinct pH and organic matter content over 30 days. While conventionally used Zn salts leached from the soil resulting in very low CaCl_2 -extractable Zn concentration, Zn in ZnO NPs was less labile, and ZnO biopolymers maintained a better constant supply of CaCl_2 -extractable Zn than all other treatments. ZnO NPs/alginate beads prepared by cross-linking with CaCl_2 presented the slowest Zn release kinetics. As assessed with maize plants grown in poor Zn acidic soil (LUF 2.1, pH 5.2), this constant Zn release from ZnO NPs/alginate beads resulted in a steadier Zn concentration in soil pore water over time. Results further indicated that ZnO NPs/alginate beads could meet the maize Zn needs while avoiding the early stage Zn toxicity induced by conventional Zn supplies, demonstrating that these ENMs are a sustainable way to supply Zn in a controlled manner in acidic soils. The impact of plant exudates on Zn bioavailability in the soil under maize-root influence (rhizosphere) is also discussed, underlying the need to study the fate of micronutrients in the rhizosphere to better predict its long-term bioavailability in bulk soils.

KEYWORDS: controlled release, micronutrients, biopolymers, composites, zinc oxide nanoparticles, maize rhizosphere



1. INTRODUCTION

In the last decades, nanotechnology has achieved great progress, and it has been applied with success in several fields such as electronics, energy, environmental science, and medicine.^{1–4} Nanotechnology applications in agriculture are increasing and need further exploration.⁵ Indeed, there are already some examples in the literature that show the potential of nanomaterials in improving agrochemical delivery, seed germination, plant growth and protection, pathogen detection, and pesticide residue detection.^{6–12}

Recently, nanomaterials were used in agricultural production to mitigate environmental problems caused by conventional intensive application of fertilizers.^{13,14} Because of a lack of synchronization between the release of minerals (often away from the roots) from bulk ionic fertilizers and uptake by plants, only a small part of the fertilizers applied to the soils are actually used by plants (30–50%).¹⁵ The remaining zinc (Zn) applied forms nonbioavailable complexes in soils, or it is removed by leaching or runoff, causing negative environmental impacts.^{15–17} Using nanomaterials to apply essential micro- and macronutrients to plants is a promising fertilization method.¹⁸ Nanomaterials can be used to control the release of

nutrients in soil/plant systems according to specific soil biochemical conditions, plant species, and plant life cycle stages, allowing a reduction of nutrient loss and of fertilizer application rates.^{16,19}

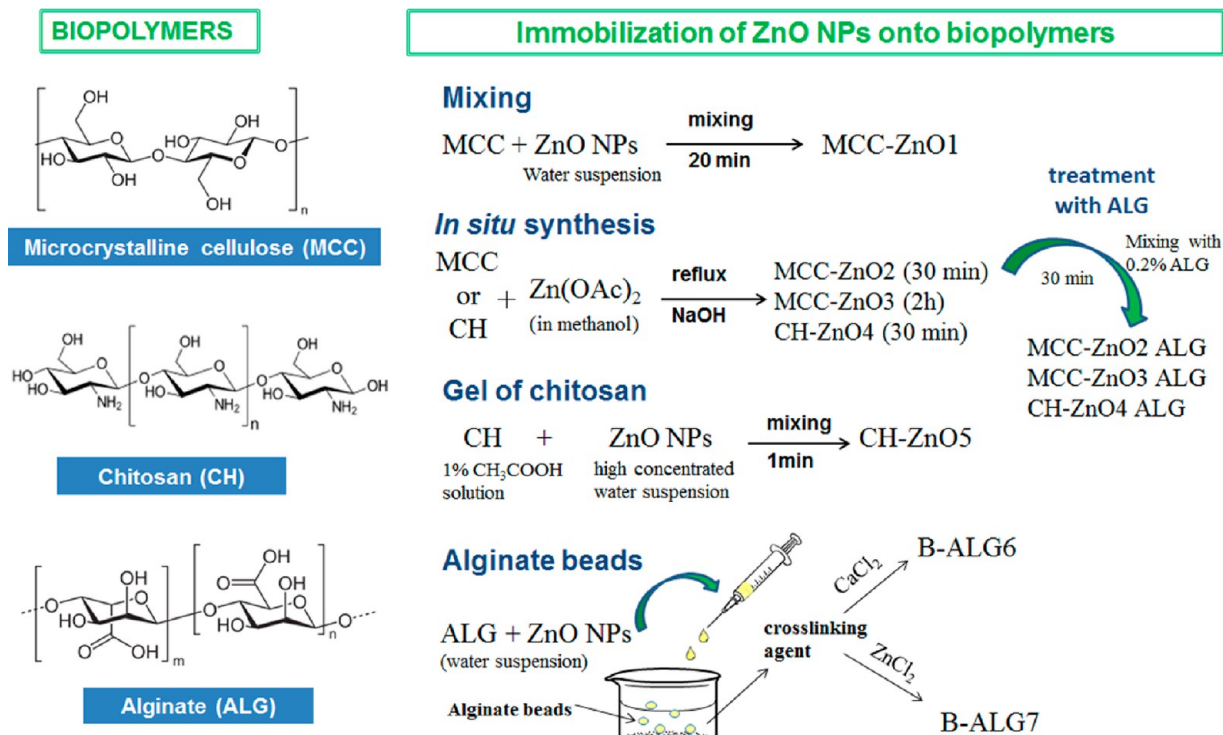
Essential plant nutrients include nitrogen, phosphate, potassium, and calcium (macronutrients) and iron, zinc, and copper (micronutrients). Zn is crucial for plant growth and development because it takes part in a wide range of plant biochemical processes. Zn plays a vital role in chlorophyll synthesis, the regulation of plant growth hormones, the maintenance of healthy root systems, the activation of enzymes and detoxification of free radicals, and the preservation of tolerance to plant stressors.^{18,20} This element is also a very important nutrient for human health. In developing countries, where staple diets are plant-based, Zn deficiency is associated with growth retardation, impaired brain

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Scheme 1. Methods Used for the Preparation of Composites Prepared in This Work



development, and increased susceptibility to infectious diseases such as pneumonia and diarrhea.²¹

Wheat, rice, and maize are strategic crop plants for human nutrition and very sensitive to Zn deficiencies in soils.²² To grow without developing signs of Zn deficiencies, gramineous plants, and maize in particular, usually need around 1 mg of extractable Zn per 1 kg of soil,²³ or 50 μg of Zn per 1 L of soil pore water.¹⁴ This bioavailable Zn concentration will depend on the soil pH, oxide species, and organic matter content.^{24,25}

Zn deficiency is a common problem in soils worldwide.²⁶ To correct Zn deficiency and prevent losses in crops, fertilizers containing Zn are added to the soils. Water-soluble salts such as ZnSO₄ or ZnCl₂ are often used as Zn fertilizers.^{27,28} These salts are a source of Zn readily available for plants; however, these nutrients can be easily lost by leaching and runoff from soils, particularly in acidic soils under heavy rainfall or intensive watering. ZnO in bulk or nanoparticle (NP) form can be an alternative to the use of ionic Zn forms as fertilizers. The effect of ZnO NPs on soils and crops has recently been the subject of debate. There are a few papers reporting the phytotoxicity of ZnO NPs; however, in the majority of these studies, the experiments were designed using very high doses of ZnO NPs (higher than those used in conventional fertilization) and very short exposure times.^{29,30} In fact, there are also many studies that show that, at low doses, ZnO NPs are beneficial to crops.^{14,16,31} Moreover, a recent study³² demonstrated that in acidic soils, because of the rapid dissolution of ZnO NPs, the risks of using ZnO NPs as fertilizers are not different from those of using ionic Zn.

ZnO NPs dissolve relatively fast in acidic soils and can also be subject to aggregation, leading to low Zn availability to crops and significant Zn losses by soil leaching and wash-off, which result in Zn contamination of surface and ground-water.^{13,32} We hypothesize that the immobilization of ZnO NPs onto polymeric substrates to form composites that release

Zn in a slower and controlled manner can be a solution to these fertilization's shortcomings.

Biopolymers such as cellulose, alginate, or chitosan (CH) can be interesting choices to be used as substrates for the immobilization of NPs to produce composite fertilizers because these are natural, low-cost, and biodegradable polymers.³³ Moreover, some of these polymers are themselves beneficial to plants. This is the case of CH, which is known to act as a promoter of plant growth and as an activator of plant defense system.³⁴ There are reports in literature concerning the use of biopolymers for the controlled release of nutrients; however, the majority of the research is directed to macronutrients, and the release tests are mainly done in water.^{35–37} Soil is a very complex matrix, and the nature and composition of the soil have a significant impact on the behavior and fate of the materials, particularly on dissolution. Tests of the release of nutrients from polymer-based materials in realistic soil conditions are therefore most needed.

In this work, a series of composites of biopolymers and ZnO NPs were prepared (by the methods shown in Scheme 1), their potential as materials for the controlled release of Zn in agricultural soils was tested, and their impact toward maize plant was assessed. The materials were added to soils with distinct pH and organic matter content. A screening soil incubation study (7 days) was performed to determine the amount of labile Zn dissolved from the different ZnO NPs composites along time in different agricultural soils, using a soil 0.01 M CaCl₂ extraction.³⁸ The rate of Zn release was correlated with the type of polymeric matrix, the method of preparation of the materials, and the characteristics of the soils. The extractability of Zn from the most promising material, ZnONPs/alginate beads, was further studied in an experiment with a soil incubation period of 30 days to evaluate the potential of this material to be used for the controlled release of Zn in agricultural soils. Finally, the ability of ZnO NPs/

Table 1. List of All Composites Prepared

sample	matrix	method	ZnO NPs diameter ^a (nm)	Zn ^b (% w/w)	deposition efficiency (%)
MCC-ZnO1	MCC	mixing MCC with ZnO NPs	87 ± 24 ^c	3.6 ± 0.6	54 ^d
MCC-ZnO2	MCC	in situ synthesis of ZnO NPs, 30 min of reaction	129 ± 39	13.5 ± 2.6	85 ^e
MCC-ZnO2 ALG	MCC	treatment of MCC-ZnO2 with alginate	129 ± 39	12.9 ± 2.5	
MCC-ZnO3	MCC	in situ synthesis of ZnO NPs, 2 h of reaction	120–400	15.1 ± 1.8	94 ^e
MCC-ZnO3 ALG	MCC	treatment of MCC-ZnO3 with alginate	120–400	13.2 ± 1.5	
CH-ZnO4	CH	in situ synthesis of ZnO NPs, 30 min of reaction	113 ± 33	12.9 ± 1.6	81 ^e
G-CH-ZnO5	CH	gel of CH and ZnO NPs	87 ± 24 ^c	64.6 ± 7.0	87 ^d
B-ALG6	ALG	ZnO NPs/alginate beads cross-linking with CaCl ₂	87 ± 24 ^c	11.8 ± 1.1	88 ^d
B-ALG7	ALG	ZnO NPs/alginate beads cross-linking with ZnCl ₂	87 ± 24 ^c	33.5 ± 3.5	>100 ^d

^aBased on SEM images of at least 150 particles (the results are presented as mean ± standard deviation or size range). Only two composites (MCC-ZnO3 and MCC-ZnO3 ALG) exhibited significantly higher diameter ($p < 0.05$) than the remaining ones. ^bThe results are presented as mean ± standard deviation (three replicates of each material). ^cCommercial NPs. ^dDeposition efficiency = ratio between the mean mass of Zn measured in the composite and the mass of Zn in the suspension of ZnO NPs. ^eDeposition efficiency = ratio between the mean mass of Zn measured in the composite and the expected mass of Zn (considering a yield of 100% in the in situ synthesis of ZnO NPs).

alginate beads to supply Zn to maize plants was tested in soil LUFA 2.1, in comparison to ZnO NPs and ZnCl₂ for 30 days. We hypothesized that this material can potentially be used to control the effectivity of the application of Zn fertilizer in soil and thus reduce environmental collateral damages of Zn loss upon soil application.

2. EXPERIMENTAL SECTION

2.1. Materials and Analytical Methods. Microcrystalline cellulose (MCC), chitosan (CH; low molecular weight, degree of deacetylation of 75–85%), and alginic acid sodium salt from brown algae (ALG) were purchased from Sigma-Aldrich and used as received. Calcium chloride (>99%, ACS grade), zinc chloride (>98%, ACS grade), and zinc acetate dihydrate (>98%) were obtained by Merck. Commercial ZnO NPs were purchased from Nanostructure & Amorphous Materials Inc., USA (99.5% purity, 20-nm-diameter size as reported by the manufacturer).

Four soils were used in this study: standard agricultural soils LUFA 2.1 and LUFA 2.2 were purchased from LUFA Speyer in Germany; two agricultural calcareous soils were collected in south Portugal (soil codes “DE” and “VV”; soils were collected at the 0–20 cm layer using plastic spades and transported to the laboratory for pretreatment and analysis). The LUFA soils were included because they are commonly used in metal bioavailability studies, and therefore it would be easier to compare our results with others found in the literature. Soil characterization was performed on the <2 mm air-dried soil fraction. The pH of the samples was measured using a glass electrode in a 1:5 (v/v) suspension of ashes/soils in CaCl₂ (according to ISO 10390:1994). A laser diffraction particle size analyzer (Coulter LS230) was used to determine the clay, silt, and sand contents for ashes and soils (limit of detection = 0.040 μm). Organic carbon (OrgC) in soils was determined after the addition of hydrochloric acid to remove carbonates (ISO 10694:1995). The water holding capacity (WHC) was determined by saturating the air-dried soils in a cylinder for 24 h. The cylinder was then placed on an absorbent membrane until excess water was drawn away by gravity. Once equilibrium was reached (24–48 h), the WHC was calculated based on the weight of the water held in the sample versus the sample dry weight. The pseudototal content of Zn in soils was determined by aqua regia digestion (according to ISO 11466:1995) and analysis by inductively coupled plasma mass spectrometry (ICP-MS; Agilent 7700). Soil-extractable Zn was determined by extraction with 0.01 M CaCl₂ [1:10 (w/v) ratio] and analysis by ICP-MS. Samples for analysis by ICP-MS were acidified with 20% HNO₃ (final HNO₃ concentration of 1%) kept at 4 °C, and analyzed within 1–4 days after collection. Instrument calibration standards were prepared in the same acid matrix as the soil extracts, and spike recovery samples were included in all analyses. Before analysis by ICP-MS, the samples were

diluted in ultrapure water [1:10 (v/v)]. The detection limit for this Zn analysis was 8 μg/L. All analyses were performed in triplicate.

LUFA 2.1 is a loamy sand soil containing 3 wt % clay, 11 wt % silt, and 86 wt % sand. It has an OrgC content of 0.71 ± 0.08%, a pH (0.01 M CaCl₂) of 5.2, and a maximum WHC of 32%. LUFA 2.2 is a sandy loam soil containing 8 wt % clay, 15.8 wt % silt, and 76.2 wt % sand. It has an OrgC content of 1.61 ± 0.15%, a pH (0.01 M CaCl₂) of 6.1, and a maximum WHC of 43%. Moreover, these soils have a low content of extractable Zn (0.86 mg/kg for LUFA 2.1 and 0.26 mg/kg for LUFA 2.2, CaCl₂ extraction), making background interference minimal.

Soil DE has an OrgC content of 7.2%, a pH (0.01 M CaCl₂) of 7.0, and a maximum WHC of 48% and contains 0.1 wt % clay. Soil VV has an OrgC content of 0.9%, a pH (0.01 M CaCl₂) of 6.9, and a maximum WHC of 27% and contains 10.4 wt % clay. The concentration of extractable Zn in soils DE and VV was under the detection limit of ICP-MS (8 μg/L).

All of the dry soils (without Zn amendments) were sterilized by autoclaving (121 °C) 1 h prior to use in the incubation tests.

All chemicals were of analytical grade or better, and all solutions were prepared using ultrapure water (Milli-Q water).

2.2. Preparation of Nanocomposites. Table 1 presents a list of the composites prepared in this work along with the method used for their synthesis and the respective Zn content [as determined by atomic absorption spectroscopy (AAS) following digestion]. The next sections present a detailed description of the methods of preparation of the composites.

2.2.1. Composite Obtained by Mixing MCC with ZnO NPs. Commercial ZnO NPs (0.018 g) were dispersed in 50 mL of Milli-Q water, and the resulting suspension was sonicated using an ultrasonic bath for 5 min. The suspension was placed under magnetic stirring, and then 0.2 g of MCC was added. After 20 min of mixing, the resulting nanocomposite (identified as MCC-ZnO1) was collected by filtration, thoroughly washed with distilled water, and dried under vacuum in a desiccator with silica gel overnight.

2.2.2. In Situ Synthesis of ZnO NPs on Biopolymers. ZnO NPs were grown at polymer surfaces (MCC or CH) by zinc acetate hydrolysis with NaOH in an alcoholic medium in the presence of water. The method used was an adaptation of the procedure described by Wu et al.³⁹ for the synthesis of ZnO NPs. In the present study, the reaction was performed in the presence of biopolymers. In this preparation, 0.5 g of biopolymer was mixed with 40 mL of a methanolic solution of Zn(OAc)₂·2H₂O (3.8×10^{-2} M) at room temperature for 1 h. The suspension was then refluxed for 45 min under stirring, and 10 mL of water was added, followed by the dropwise addition of 30 mL of a NaOH methanolic solution (0.1 M). After 30 min or 2 h of reflux, the reaction was stopped, and the suspension was allowed to cool to room temperature. The final composites were isolated by filtration, washed two times with 10 mL of ethanol, and dried overnight under vacuum in a desiccator with

silica gel. This procedure generated the composites here identified as MCC-ZnO2 (30 min reaction), MCC-ZnO3 (2 h reaction), and CH-ZnO4 (30 min reaction).

2.2.3. Alginate-Treated MCC-ZnO Nanocomposites. Composites MCC-ZnO2 and MCC-ZnO3, obtained by in situ synthesis, were further treated with a solution of alginate. Typically, 0.2 g of MCC-ZnO2 or MCC-ZnO3 composites was treated with 50 mL of a solution containing 0.2 g of sodium alginate with stirring for 1 h. The resultant composites, identified as MCC-ZnO2 ALG and MCC-ZnO3 ALG, were then isolated by filtration, thoroughly washed with distilled water, and dried under vacuum in a desiccator with silica gel overnight.

2.2.4. Gel of CH and ZnO NPs. A gel of CH and ZnO NPs was prepared by adaptation of the method reported by Vasile et al.⁴⁰ CH (0.025 g) was dissolved in 10 mL of an acetic acid solution (1%). Commercial ZnO NPs (0.3 g) were suspended in 5 mL of distilled water and sonicated using an ultrasonic bath for 5 min. The suspension was quickly added to a 10 mL solution of CH (0.025 g) in acetic acid (1%) under magnetic stirring. In a few seconds a ZnO/CH gel was formed. The gel with a consistency of gelatin pudding was removed from the beaker and cut into pieces that were dried at room temperature. The resulting films were ground in a mortar to obtain a powder (G-CHZnO5).

2.2.5. Alginate Beads Containing ZnO NPs. Sodium alginate (0.75 g) was stirred in 50 mL of distilled water for 1 h to ensure the formation of a homogeneous solution. ZnO NPs (0.150 g) were then added to the alginate solution, and the mixture was left under continuous stirring. The milky mixture was transferred to a syringe and added drop by drop into a beaker containing 100 mL of a 0.25 M solution of the cross-linking agent (CaCl_2 or ZnCl_2) under moderate stirring. The resulting alginate beads were allowed to stay for 30 min in the cross-linking agent solution and then separated by filtration, thoroughly washed with distilled water, and dried at 30 °C for 1 day. For soil tests, the beads were ground in a mortar to obtain a powder. By this method, composites B-ALG6 (cross-linking with CaCl_2) and B-ALG7 (cross-linking with ZnCl_2) were obtained.

2.3. Materials Characterization. Scanning electron microscopy (SEM) images were obtained with a Hitachi S4100 or a Hitachi SU-70 instrument fitted with an energy-dispersive spectroscopy (EDS) accessory (EDS detector, Bruker AXS; software, *Quantax*). Samples were deposited on carbon tape and coated with carbon before SEM analysis. Transmission electron microscopy (TEM) was performed with a Hitachi H-9000 or a JEOL 2200FS electron microscope equipped with an Oxford EDS detector and an in-column Omega filter, operated at 300 kV. The samples for TEM were prepared by depositing a drop of a diluted suspension of the materials in ultrapure water onto the surface of an Agar Scientific carbon-coated copper grid and then allowing the solvent to evaporate. To calculate the mean size of ZnO NPs based on the SEM or TEM images, at least 150 particles were measured.

The total Zn concentration in the composite materials was evaluated by AAS using a Thermo Scientific ICE 3000 series AA spectrometer (detection limit: 50 $\mu\text{g/L}$). Before AAS, the samples were submitted to acid digestion with HNO_3 (68%) in a microwave oven. Instrument calibration standards were prepared in the same acid matrix as the soil extracts. All analyses were performed in triplicate.

ζ -potential measurements were performed using a Zeta Sizer Nano Series (Malvern) equipment. The optical spectra were recorded using a Jasco V-560 UV-vis spectrophotometer; for the solid samples, the spectra were recorded in the diffuse-reflectance mode using MgO as the reference. Powder X-ray diffraction (XRD) patterns of the solid samples were recorded with a Philips X'Pert instrument operating with $\text{Cu K}\alpha$ radiation ($\lambda = 1.543178 \text{ \AA}$) at 40 kV and 50 mA. Fourier transform infrared (FTIR) spectra (128 scans at a resolution of 4 cm^{-1}) were collected with a Mattson 7000 spectrometer coupled to a horizontal attenuated-total-reflectance (ATR) cell.

2.4. Soil Amendment and Incubation. Soils LUFA 2.1, LUFA 2.2, DE, and VV were amended with ZnO NPs, ZnCl_2 , or biopolymer composites loaded with ZnONPs to obtain a Zn concentration of 100 mg/kg dry soil.

ZnO NPs and ZnO NPs composite powders were dispersed in Milli-Q water (pH 7), and the suspensions were then sonicated in an ultrasonic bath for 2 min. The dispersed nanomaterials were then immediately added to the soil. ZnCl_2 powder was dissolved in Milli-Q water before being amended to the soil. The amount of Milli-Q water added to each soil was that necessary to ensure that each soil was at 50% of their maximum WHC after amendment.

First a screening test was done in acid soils LUFA 2.1 and LUFA 2.2 with all of the materials for two incubation times, 1 and 7 days. Typically, 4 or 5.6 mL of a ZnCl_2 solution or a ZnO NPs materials suspension was added to 25 g of dried LUFA 2.1 or LUFA 2.2 soils, respectively. A total of 18 g (dry weight) of each amended soil was then divided into six centrifuge tubes (three replicates and two incubation periods) with 3 g of soil each. The soil samples were then incubated under aerobic conditions at room temperature for 1 and 7 days before being extracted. The moisture content of the soil samples was kept at 50% of their maximum WHC values by daily additions of Milli-Q water.

The material that presented the lowest Zn release (B-ALG6) was tested in a period of 30 days of incubation in acid soils LUFA 2.1 and LUFA 2.2 and neutral soils (DE and VV) in order to better understand the role of pH on Zn release from this composite. In this case, soil samples were incubated under aerobic conditions at room temperature for 1, 2, 4, 7, 15, and 30 days before being extracted.

2.5. Maize Culture in LUFA 2.1 Soil Amended with Different Zn Sources. Maize seeds (*Zea mays*) were surface-sterilized with bleach, thoroughly rinsed with deionized water, and germinated on a plate with saturated humidity in the dark at 20 °C for 5 days. LUFA 2.1 soil (900 g) was amended with different treatments, as described above. Homogenization of the soil with different treatments was done by stirring the soil (3 times during 5 min) after the addition of materials suspensions to ensure good homogenization of the treatments. Soils were prepared to reach 50% of the maximum WHC values. After homogenization, the soils of different treatments were transferred into pots. In the middle of each pot, a filtration device was horizontally placed. These samplers (Rhizon Flex, 5 cm)⁴¹ are made of poly(ether sulfone) membranes of 0.12–0.17 μm nominal pore size, connected to a luer lock through a 30-cm-long poly(vinylpyrrolidone) (PVP)/polyethylene (PE) tubing, which allowed for pore water sampling outside the pot, without disturbing the soil. These membrane filters were made of inert polymers with no ion-exchange properties to minimize sorption. Pore water (4–7 mL) was sampled from each pot 2, 6, 10, 24, and 29 days after the seed transfer. An aliquot was used for pH measurement, and the rest was acidified (2% HNO_3) and stored at 4 °C for total Zn measurement by ICP-MS, as described in section 2.1.

A rope made of 100% cotton connecting the soil to a nutritive solution allowed, through capillary exchanges, to keep the soil humidity constant over time. The nutritive solution used was a $1/4$ strength Hoagland solution,⁴² prepared without Zn micronutrients to ensure that the plant will not undergo other deficiencies than Zn.

Four maize seedlings were transferred per pot of 900 g of soil amended with 30 or 100 ppm (as Zn) of either ZnCl_2 , ZnO NPs or ZnO NPs/alginate (B-ALG6), or in a control soil (ultrapure water). Alginate controls with alginate concentration similar to that of 30 and 100 ppm B-ALG6 treatments were also run. These resulted in nine treatments and four maize plants per treatment, namely, ultrapure water, ALG 30eq, ALG 100eq, ZnCl_2 30 ppm, ZnCl_2 100 ppm, ZnO 30 ppm, ZnO 100 ppm, B-ALG6 30 ppm, and B-ALG6 100 ppm. The plants were then grown under Tubo T8 G13 58W 5000LM 6500K fluorescence lamps under a 16:8 light/night photoperiod over 29 days.

At the end of the experiment, all of the plants were collected. Shoots were separated from the root system, and pictures were taken. Shoots and roots were dried at 60 °C for 48 h to assess for dry biomass. The lengths of the leaves were measured on the plant pictures using *ImageJ* software. The shoot density was calculated as a ratio of the dry biomass divided by the total leaf length of each plant.

Bulk soil was separated from the rhizosphere soil following previously established procedures,⁴³ dried at 60 °C until the dry mass

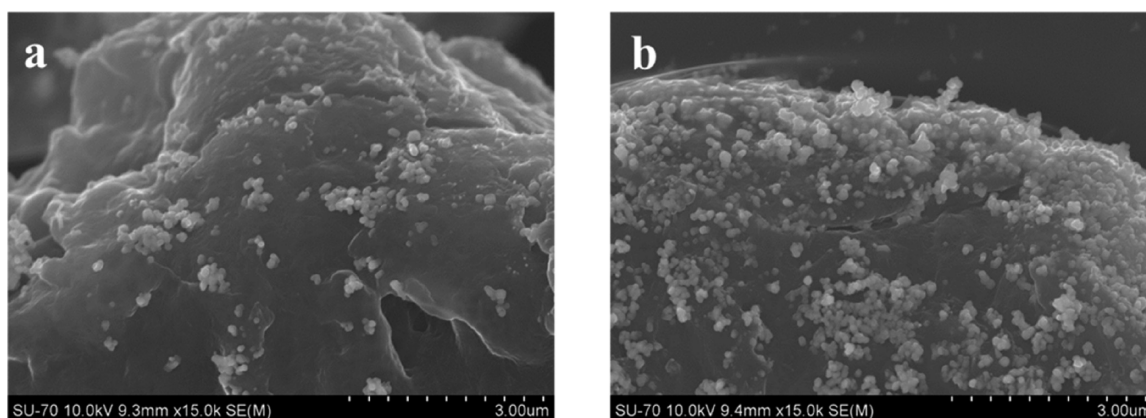


Figure 1. SEM images of the MCC-ZnO1 composite showing areas with low (a) and high (b) densities of ZnO NPs.

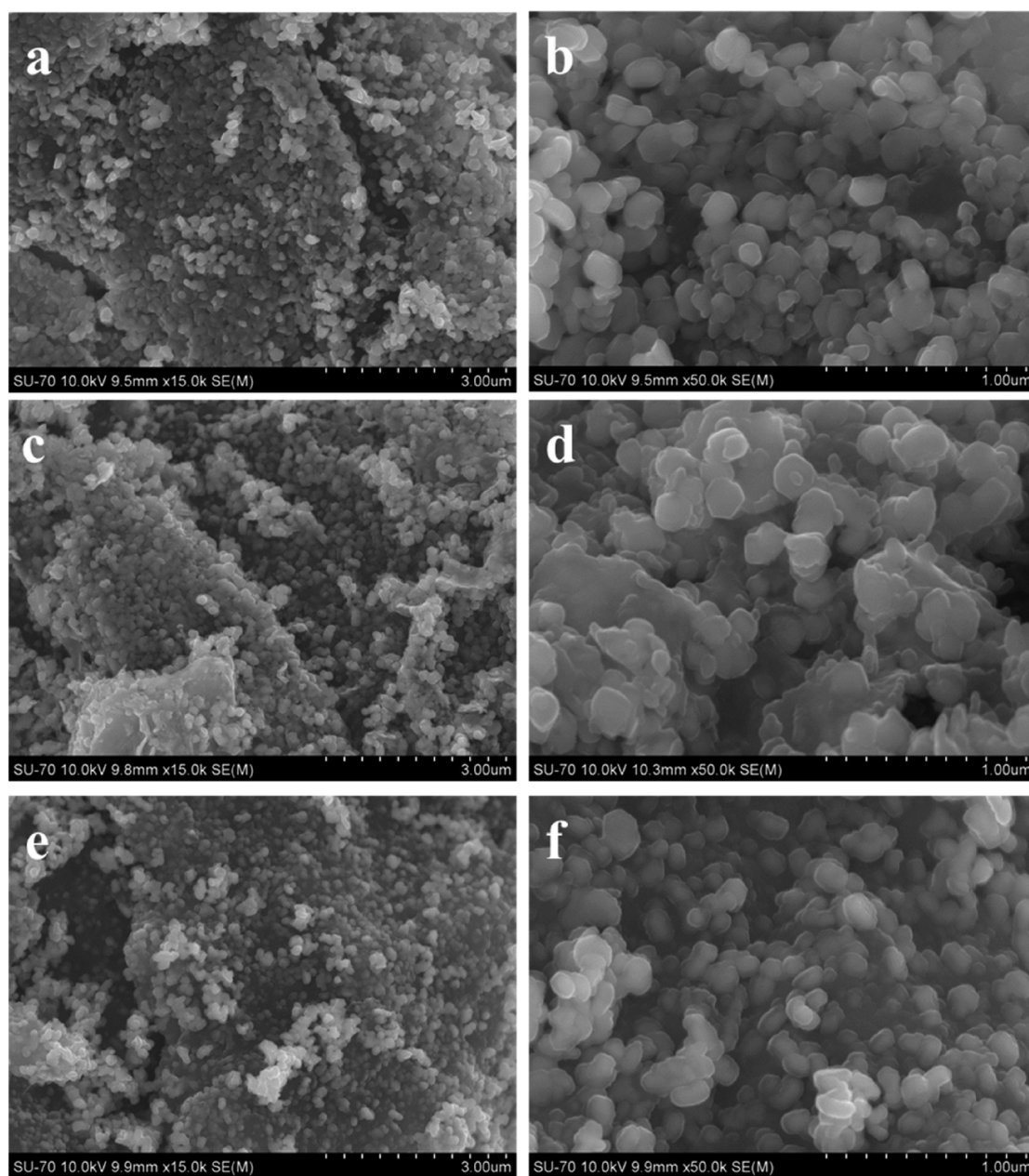


Figure 2. SEM images of MCC-ZnO₂ (a and b; 30 min reaction), MCC-ZnO₃ (c and d; 2 h reaction), and CH-ZnO₄ (e and f; 30 min reaction) nanocomposites.

did not change, and used for CaCl_2 extraction, as described below for the pH and Zn concentration measurements.

2.6. Extraction Test To Determine the Extractable Pool of Zn in Amended Soil. After 1, 2, 4, 7, 15, or 30 days of incubation without plants and 29 days of incubation with plants, dry soil samples of soils LUFA 2.1, LUFA 2.2, DE, and VV amended with ZnO NPs, ZnCl_2 , or biopolymer composites loaded with ZnONPs were extracted in triplicate with a 1:10 soil/ CaCl_2 ratio using a 0.01 M CaCl_2 solution in centrifuge tubes. The centrifuge tubes were laid horizontally and shaken for 2 h, using a reciprocal shaker at 180 rpm. After extraction, all samples were centrifuged at 1022 g for 10 min, and the supernatants were filtered using a 0.2 μm poly-(tetrafluoroethylene) syringe filter. It should be noticed that at first supernatants were filtered with 0.2 μm and 3 kDa filters, and it was observed that the Zn contents of the extracts were similar in the two cases. Therefore, in the following experiments, only 0.2 μm filters were used. Soil blanks (no Zn treatment; only ultrapure water was added to ensure the same moisture content) were used as the controls. The pH values of the CaCl_2 extracts were measured. After that, all extracts were acidified with a 20% solution of HNO_3 (final HNO_3 concentration of 2%) and analyzed for the Zn concentration by ICP-MS, as described in section 2.1.

2.7. Statistical Analysis. The software SPSS 24.0 by IBM for MacOS was used for calculation of the descriptive statistics and for statistical analysis of the data. For the experiment without maize, a comparison of results for different materials and respective Zn releases in soil was based on the analysis of variance (ANOVA) provided by the General Linear Model (univariate analysis). Bonferroni post hoc tests were applied to explore the differences between the group means. To compare the plant biomass and density, ANOVA was done using a Tukey test.

3. RESULTS AND DISCUSSION

3.1. Materials Characterization. SEM analysis of commercial ZnO NPs powders showed that the particles were nearly spheroidal with a mean diameter of 87 ± 24 nm (Figure S1). TEM analysis of a diluted suspension of ZnO NPs in water was also performed (Figure S2). The diameter of the NPs calculated from the TEM images was in good agreement with that obtained by SEM, which also showed that the ZnO NPs did not aggregate in water (at least in the time of preparation of the suspension: 5 min). The size distribution of ZnO NPs calculated from the SEM and TEM images was higher than that reported by the manufacturer (20 nm). This discrepancy could be attributed to some aggregation of metal oxide NPs or to the low accuracy of the size values reported.⁴⁴ ζ -potential measurements of a 200 mg/L ZnO NPs suspension showed that the NPs are positively charged ($+22.1 \pm 0.379$ mV at pH 7.2). The powder XRD pattern of ZnO NPs (Figure S3) presents the characteristic peaks of a wurtzite crystal structure (JCPDF 36-1451). The UV-vis reflectance spectrum of ZnO NPs showed the characteristic absorption edge associated with the energy gap of ZnO, with the maximum of absorption at 360 nm (Figure S4). The absorption maximum of ZnO NPs was blue-shifted relative to bulk ZnO (380 nm) because of the quantum confinement effect of the ZnO nanostructure.⁴⁵

Several composites of biopolymers (MCC, CH, and alginate) and ZnO NPs were produced using different methodologies (Table 1 and the Experimental Section).

The MCC-ZnO1 composite was prepared by mixing MCC with commercial NPs. The efficiency of the deposition of ZnO NPs on MCC fibers was low (54%), and the Zn content on the final composite was only 3.6% (w/w). The low deposition efficiency could be related with the fact that positively charged ZnO NPs can only bind to negatively charged MCC fibers (ζ

potential of -18.6 mV at pH 7.2) by weak electrostatic forces. SEM (Figure 1), EDS (Figure S5), and XRD (Figure S6) analyses of the MCC-ZnO1 sample confirmed the presence of ZnO NPs over MCC fibers; however, the NPs do not fully cover the polymer surface. Moreover, deposition of the ZnO NPs seems to be nonhomogeneous because it is possible to find areas in the matrix with low (Figure 1a) and high (Figure 1b) densities of NPs. Areas with a high density of NPs were attributed to the existence of a higher number of carboxyl groups (which have a negative charge) on the MCC fibers. XRD of the sample (Figure S6) shows the characteristic Bragg reflections of ZnO (wurtzite; $2\theta = 31.8^\circ, 34.6^\circ, 36.4^\circ, 47.8^\circ, 56.7^\circ, 63.1^\circ$, and 68.3°) and of MCC (cellulose I; $2\theta = 14.3^\circ, 15.9^\circ$, and 22.6°). The UV-vis reflectance spectrum of the nanocomposite showed the presence of the absorption edge associated with the energy gap of ZnO, with an onset of absorption close to that observed in the spectrum of commercial ZnO NPs (Figure S4).

In situ synthesis of ZnO NPs performed in the presence of biopolymers generated composites with a higher content of Zn than those prepared by mixing the biopolymer with ZnO NPs (MCC-ZnO1). Using different biopolymers (MCC or CH) and variable reaction times (30 min or 2 h), three different composites were obtained with in situ synthesis of ZnO NPs (MCC-ZnO2, MCC-ZnO3, and CH-ZnO4). SEM analysis of the nanocomposites showed that the NPs tended to agglomerate at the surface of the polymer, forming a compact ZnO NPs coating (Figure 2). Unlike the commercial NPs used in this work, which are nearly spherical in shape, the ZnO NPs obtained by in situ synthesis had a disk shape, with some of the NPs showing hexagonal facets. For the same time of reaction (30 min), in situ synthesis of ZnO NPs onto MCC or CH occurred in a similar way. ZnO nanodisks with similar diameters were obtained (129 ± 39 nm for MCC and 113 ± 33 nm for CH; values represent the mean $\pm$ standard deviation), and the Zn contents of the composites were not very different (13.5 for MCC and 12.9% for CH). Moreover, the ZnO NPs deposition efficiencies for MCC-ZnO2 and CH-ZnO4 were also similar, 85% and 81%, respectively. This suggests that the type of matrix is not determinant in the in situ synthesis and growth of the ZnO NPs. By contrast, the reaction time had a relevant influence on the synthesis and deposition of ZnO NPs on the polymer. When the reaction time was increased from 30 min to 2 h, larger ZnO particles were formed and the composite obtained showed a higher Zn content (15.1%) and a higher deposition efficiency (94%). For the composites obtained by 30 min of reaction, the size of the NPs was fairly uniform. For the composites prepared by 2 h of reaction, particles of variable diameter were formed (from 120 to 400 nm), and in some areas, the formation of a continuous ZnO film was observed (Figure 2d). This means that long reaction times favor nonhomogeneous growth of NPs on the biopolymer. Although most of the NPs were found on the surface of the polymer, it was possible to observe the presence of ZnO NPs also inside the cellulose fibers through TEM analysis of the sample MCC-ZnO2 (Figure S7). This could be explained by considering that a Zn precursor can penetrate onto the fibers and then undergo hydrolysis in the presence of a NaOH solution, producing ZnO NPs. Other authors also observed the formation of NPs inside cellulose films or within regenerated cellulose fibers after the in situ synthesis of ZnO NPs.^{46,47} The XRD pattern of the samples showed peaks corresponding to the polymers and ZnO wurtzite phase,

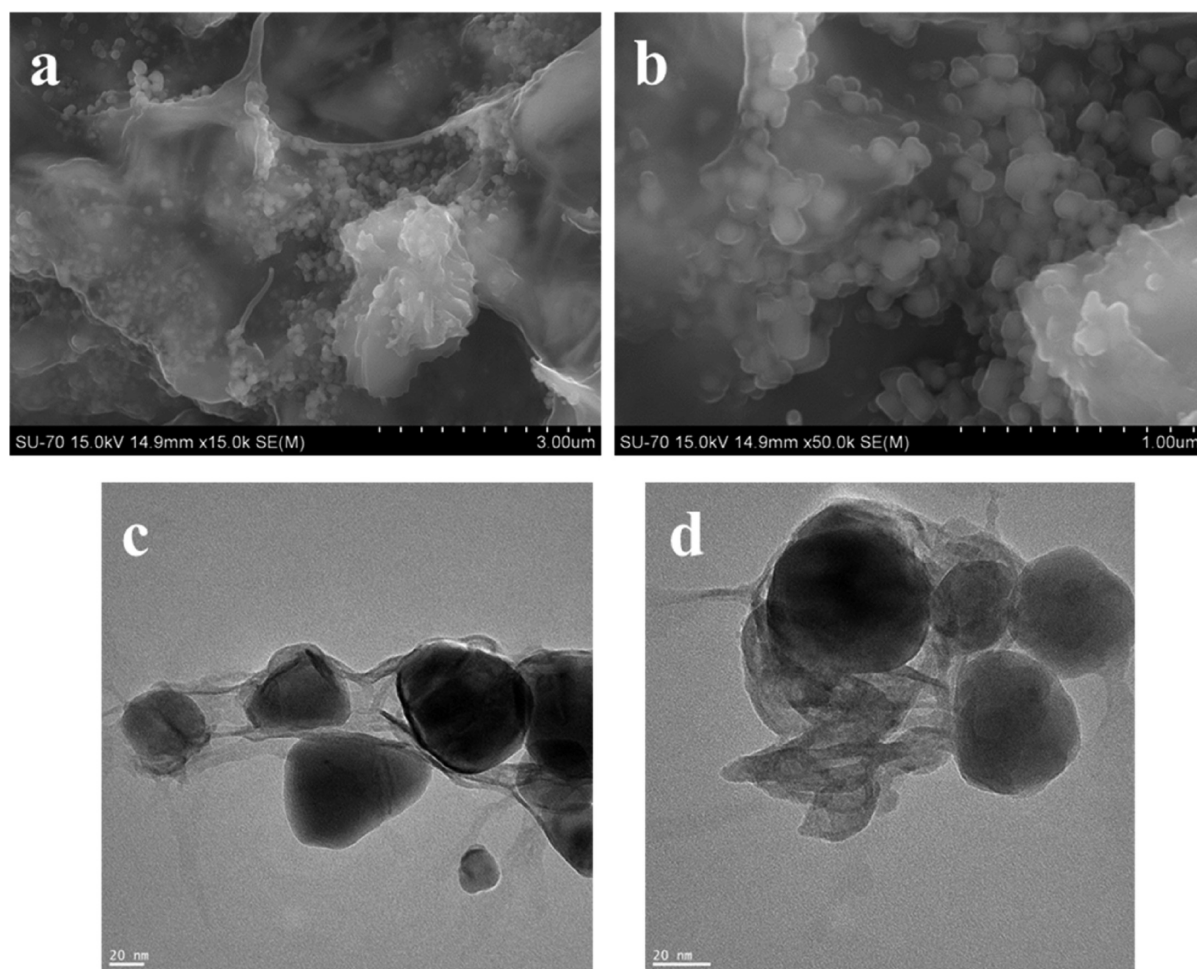


Figure 3. SEM (a and b) and TEM (c and d) images of G-CH-ZnO5.

confirming the presence of ZnO NPs on the composites (Figure S8). As expected, the UV–vis spectra of MCC-ZnO2, MCC-ZnO3, and CH-ZnO4 presented the absorption edge associated with the energy gap of ZnO (Figure S4).

The composites MCC-ZnO2 and MCC-ZnO3 were further treated with a solution of alginate. Coating the composites with a layer of alginate (with a charge opposite to that of ZnO NPs) was tested to evaluate potentially slower Zn release. By SEM analysis of the composites obtained after treatment with alginate (MCC-ZnO2 ALG and MCC-ZnO3 ALG), it was not possible to identify a layer of polymer covering ZnO NPs (Figure S9). However, ζ -potential measurements showed an increase of the surface charge of the composites obtained after treatment with alginate compared with the charge of the materials before treatment (Table 2). In the case of MCC-ZnO3 ALG, a charge reversal was even observed. This suggested that the negatively charged alginate attached to the composites surface.

Entrapment of nutrients and active ingredients on biopolymers is known to be helpful for the controlled release of these compounds.^{40,48} Having this in mind, a composite with a high content of Zn (64.6%) was prepared by the entrapment of ZnO NPs in a CH gel (G-CH-ZnO5). To prepare the composite, a suspension of commercial ZnO NPs was added to a CH solution under agitation. According to Vasile et al.,⁴⁰ CH amino groups interact with ZnO NPs, then the polymer chains wrap around the particles, and as a result, a

Table 2. ζ Potential and pH of Suspensions of MCC-ZnO Composites Treated and Not Treated with Alginate^a

sample	ζ potential (mV)	pH
MCC-ZnO2	-18.6 ± 0.7	7.30 ± 0.10
MCC-ZnO2 ALG	-32.0 ± 3.2	7.78 ± 0.20
MCC-ZnO3	$+6.0 \pm 2.5$	7.36 ± 0.20
MCC-ZnO3 ALG	-33.9 ± 2.1	7.75 ± 0.20

^aThree replicate measurements were performed (values represent the mean $\pm$ standard deviation). Both the ζ potential and pH of composites treated with alginate were significantly different ($p < 0.05$) from those of the nontreated ones.

CH gel is formed with ZnO NPs trapped inside. In fact, SEM analysis of the milled dried gel clearly showed areas where a film of CH was covering ZnO NPs (Figure 3a,b). Moreover, TEM images showed that the ZnO NPs were, in fact, wrapped by a polymer layer (Figure 3c,d). FTIR-ATR analyses also showed that there was a strong interaction between CH functional groups and the ZnO surface (Figure S10). CH displays characteristic vibrational bands at 3374 cm^{-1} (O–H stretching and N–H stretching in the NH_2 group), 2868 cm^{-1} (CH_2 , C–H symmetric stretching), 1651 cm^{-1} (stretching vibration of C=O in amide), 1585 cm^{-1} (NH_2 bending), $1050\text{--}893\text{ cm}^{-1}$ (vibrations of C–O bonds). For composite G-CH-ZnO5, the band at 3374 cm^{-1} shifted to 3364 cm^{-1} and became more intense and broad. The peak at 1585 cm^{-1} corresponding to NH_2 bending also became more intense and

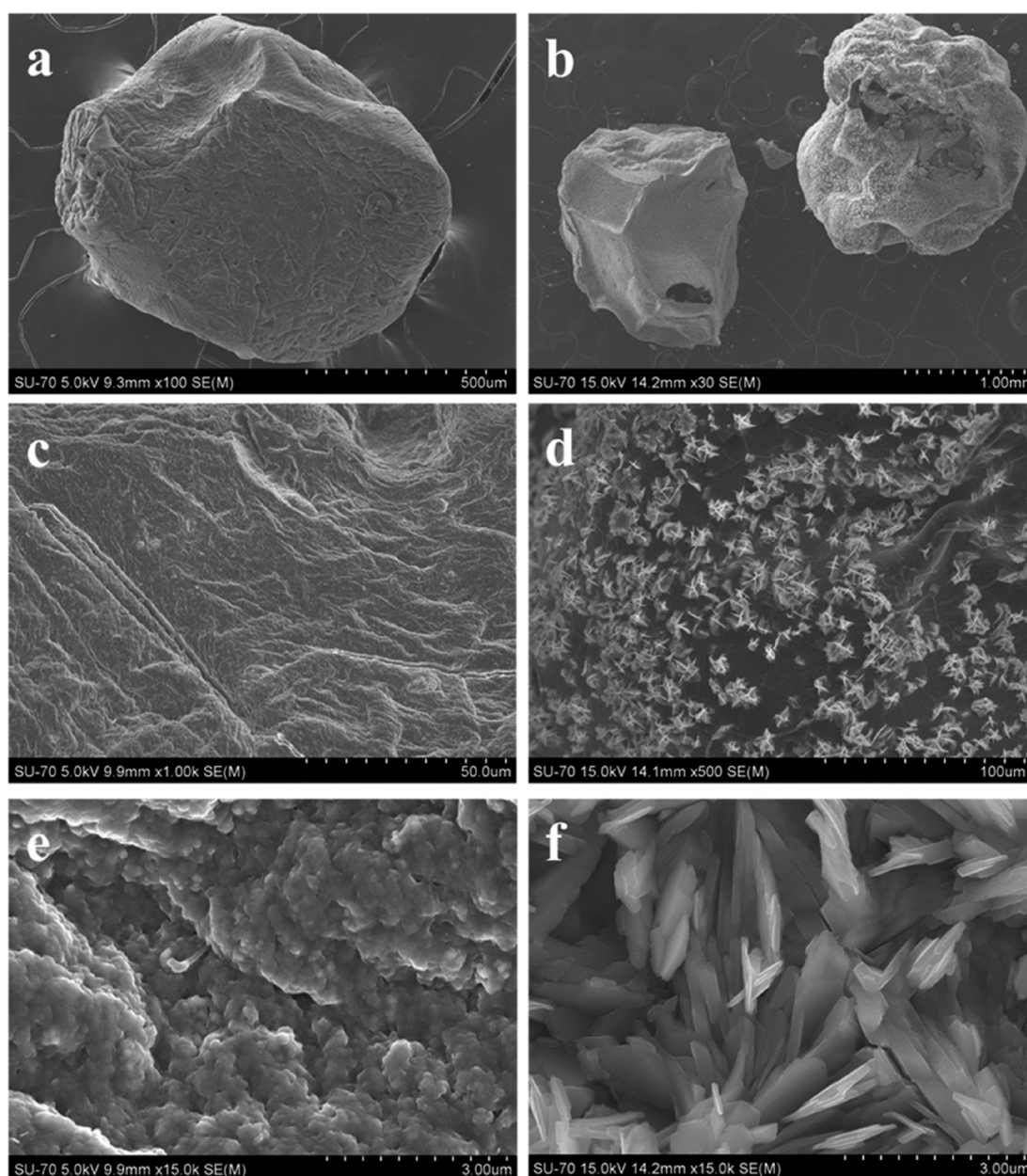


Figure 4. SEM images of B-ALG6 (a, c, and e) and B-ALG7 (b, d, and f) nanocomposites.

shifted to lower wavelength (1553 cm^{-1}). The shift and intensity increase of the FTIR peaks was attributed to the interaction of ZnO with OH and NH_2 groups of CH.⁴⁹ XRD of G-CH-ZnO5 showed the characteristic peaks of ZnO (wurtzite) as the main crystalline phase in the composite (Figure S11).

Other composites were prepared by the entrapment of ZnO NPs inside a gel-like polymeric matrix. Two types of alginate beads containing ZnO NPs were prepared by dropping a suspension of ZnO NPs onto a solution of CaCl_2 (B-ALG6) or ZnCl_2 (B-ALG7). The wet beads exhibited spherical shape with a diameter of 1–3 mm (Figure S12). After air drying, the diameter of the beads decreased to about 1 mm; however, most of particles maintained the spherical shape. The XRD pattern for both alginate beads showed the characteristic peaks of the ZnO wurtzite phase (Figure S11).

The Zn content of the B-ALG6 beads was found to be 11.8%, which means that 88% of the ZnO NPs that were in the

suspension were entrapped by the polymer. SEM analysis of dried B-ALG6 beads showed a rough surface with some wrinkles and cracks (Figure 4a,c). In the high-resolution image of B-ALG6, it was possible to observe the ZnO NPs wrapped by alginate at the bead surface (Figure 4e). Beads were cut into small pieces, allowing analysis of the interior of the beads. The SEM images and Zn EDS mapping of these pieces showed that ZnO NPs were homogeneously distributed in the composite (Figure S13).

For B-ALG7 beads, the Zn content (33.5%) was more than 2 times higher than that obtained for B-ALG6, and the calculated efficiency of the entrapment was higher than 100%. This was due to the fact that the Zn present in the composite was derived from not only ZnO NPs but also ZnCl_2 , which acted to cross-link the alginate chains. SEM analyses of the surface of the beads showed not only ZnO NPs but also abundant crystal structures (Figure 4d,f). It is possible that these crystals are ZnCl_2 , which precipitated at the surface of

the polymers, or ZnO plates obtained by the growth of ZnO NPs in the presence of ZnCl_2 . EDS mapping of Zn on a fragment of the B-ALG7 bead revealed that this element is homogeneously distributed on the polymer surface (Figure S14).

The ZnO NPs deposition efficiencies for the composites G-CH-ZnO5 and B-ALG6 were 87% and 88%, respectively. These values are similar to the deposition efficiencies of the composites obtained by an in situ method (MCC-ZnO2, MCC-ZnO3, and CH-ZnO4), which were between 81% and 94%.

Finally, the pH values of the suspensions of all of the prepared composites were measured, and values between pH 6.65 and 7.78 were obtained (Table S1). The pH values of the suspensions of MCC-ZnO composites treated with alginate (MCC-ZnO2 ALG and MCC-ZnO3 ALG) were significantly higher ($p < 0.05$) than those of suspensions of all remaining materials.

3.2. Extractability of Zn from All Biopolymer Composites in Agricultural Soils. The concentrations of 0.01 M CaCl_2 -extractable Zn for soil LUFA 2.2 amended with ZnO composites, ZnO NPs, and ZnCl_2 (100 mg of Zn per 1 kg of dry soil) after 1 or 7 days of incubation were determined (Figure 5). For all of the materials including ZnCl_2 , Zn

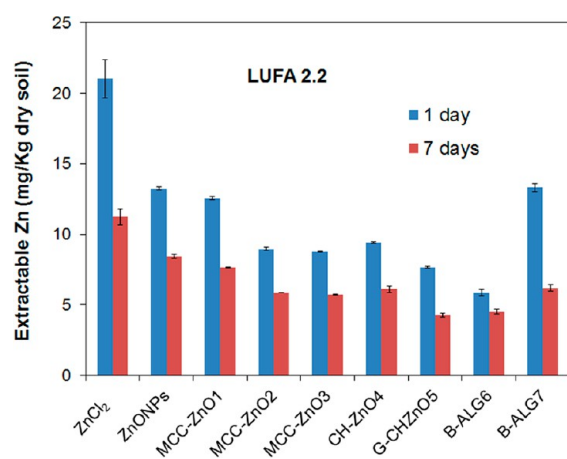


Figure 5. Extractable Zn from soil LUFA 2.2 (pH 6.3; 1.61% OrgC) amended with ZnCl_2 , ZnO NPs, and ZnO composites, incubated for 1 or 7 days. Bars represent the mean values, and error bars represent the standard deviation ($n = 3$).

extractability decreased with time. Lower Zn extractability over time in soils amended with ZnCl_2 , ZnSO_4 , or ZnO NPs was also reported by other researchers^{50,51} and is discussed further below. The extractable Zn for ZnO NPs-amended soil was always lower than that for ZnCl_2 -amended soil, suggesting that ZnO NPs were not completely dissolved after 7 days of incubation in soil. The Zn extractability in soil LUFA 2.2 amended with ZnCl_2 was significantly higher ($p < 0.05$) than those of amendments with ZnO NPs composites.

With the exception of B-ALG7, all amendments with ZnO NPs composites led to Zn extractability from soils lower than those for amendments with ZnO NPs alone (Figure 5 and Table S2). In fact, the Zn extractability decreased between (5.1% and 55.8%), suggesting a lower release of Zn on soils when ZnO NPs were immobilized in a polymeric matrix. Although slightly lower, MCC-ZnO1 presented Zn release close to those observed for the ZnO NPs alone. This was

attributed to the fact that ZnO NPs on the MCC-ZnO1 composite were mostly deposited onto the surface of MCC fibers and only attached by electrostatic forces. When in the soil, the ZnO NPs could easily detach, making the material behave similarly to that observed for nonimmobilized ZnO NPs. Composites produced by in situ synthesis (MCC-ZnO2, MCC-ZnO3, and CH-ZnO4) presented a lower Zn release (extractability was 1.2–1.4 lower) compared with that of MCC-ZnO1. In this case, it is possible that most of the ZnO NPs were attached to the polymer by chemical bonds or trapped between polymeric fibers, making Zn release more difficult. However, the release of Zn from MCC-ZnO2, MCC-ZnO3, and CH-ZnO4 was similar, showing that the type of polymeric substrate and size of the ZnO NPs had no decisive role in the Zn release for the materials prepared by an in situ method.

MCC-ZnO2 ALG and MCC-ZnO3 ALG (prepared by the treatment of MCC-ZnO composites with alginate) showed Zn release levels similar to those of the untreated composites (Table S2); however, it is possible to see a trend of lower release (2–10% less) for the treated materials.

The materials showing the slowest Zn release were those prepared by the entrapment of ZnO NPs in a gel structure (G-

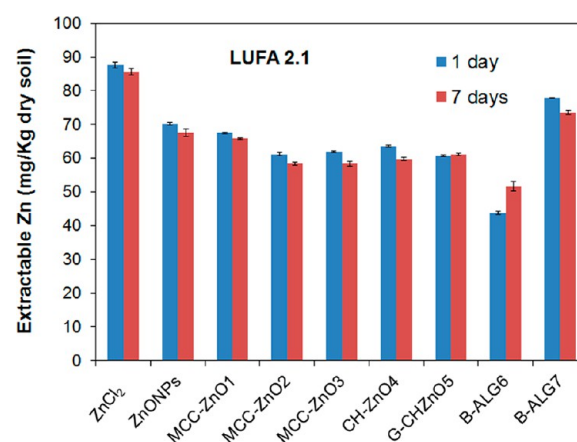


Figure 6. Extractable Zn from soil LUFA 2.1 (pH 5.2; 0.71% OrgC) amended with ZnCl_2 , ZnO NPs, and ZnO composites, incubated for 1 or 7 days. Bars represent the mean values, and error bars represent the standard deviation ($n = 3$).

CHZnO5 and B-ALG6). Clearly, the extractability of Zn from soil LUFA 2.2 amended with B-ALG6 alginate beads was 2 and 3 times lower than those of the soil amended with ZnO NPs and ZnCl_2 , respectively. Although the Zn extractability for G-CHZnO5 at the end of day 1 was higher than for B-ALG6, it became similar after 7 days of incubation. For these materials, water molecules need to diffuse through pores and channels of the polymeric gel to get in contact with ZnO NPs and to dissolve them. After that, the ionic Zn must diffuse throughout of the gel to be released onto soil pore water. This process hinders rapid dissolution of ZnO NPs.

It should be mentioned that B-ALG7 alginate beads that were produced using ZnCl_2 as the cross-linking agent instead of CaCl_2 presented almost the same Zn release after 1 day of incubation as that for the amendment with ZnO NPs alone. This could be explained by the fact that at least half of the Zn present in the composite was not in ZnO NPs form. In fact, part of the Zn was founded at the alginate chains (as the cross-linking agent) or adsorbed at the surface of the polymer.

It was clear that the fraction of extractable Zn was significantly higher in soil LUFA 2.1 (44–88% of added Zn content) than in soil LUFA 2.2 (5–21%) for all of the materials, including ZnCl₂ (Figure 6 and Table S3). This was attributed to the lower pH (5.2) and lower organic matter content (0.71% OrgC) of soil LUFA 2.1 compared with those of LUFA 2.2 (pH 6.3; 1.61% OrgC). Other authors also reported that soils with low pH and low organic matter content show a higher lability of Zn.^{52,53} The Zn extractability in soil LUFA 2.1 amended with ZnCl₂ was significantly higher ($p < 0.05$) than those of ZnO NPs composites and ZnO NPs alone.

For both LUFA 2.2 and LUFA 2.1 soils, all of the composites presented lower Zn extractability compared with that of ZnO NPs, with the exception of B-ALG7. However, the differences in Zn release between the different composites were clearly smaller for soil LUFA 2.1. This shows that the percentage and rate of Zn release from the composites is determined not only by the type of attachment of the ZnO NPs to the polymer but also by the soil properties, particularly the pH and OrgC content. It was clear that the alginate beads containing ZnO NPs prepared by cross-linking with CaCl₂ (B-ALG6) were the materials that presented the lower Zn release in both soils.

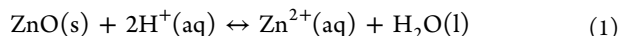
3.3. Extractability of Zn from Agricultural Soils Dosed with ZnONPs/Alginate Beads Composites over 30 Days.

A soil test with a longer incubation time (30 days) and more data points (days 1, 4, 7, 15, and 30) was performed to further evaluate the Zn release behavior from B-ALG6 over time. The behavior of B-ALG6 was compared with that observed for ZnCl₂ in the same period of time.

The concentrations of CaCl₂-extractable Zn from soils DE (pH 6.9) and VV (pH 7) amended with ZnCl₂ were between 0.736 mg/kg (day 1) and 0.09 mg/kg (day 30). For B-ALG6, the concentration of extractable Zn was below the detection limit of ICP-MS analysis from day 1 to day 30 (Table S4). This suggests that these materials may not be efficient for the controlled release of Zn in neutral/alkaline soils.

The concentrations of 0.01 M CaCl₂-extractable Zn for soil LUFA 2.2 amended with B-ALG6 and ZnCl₂ over 30 days are shown in Figure 7a. The materials exhibited distinct behavior in relation to extractable Zn. In the ZnCl₂-amended soil, the extractable Zn was high on the first day and then showed a fast decrease over time. A decrease of 60% in the extractable Zn was observed from day 1 to day 15 of incubation in soil. After 15 days of incubation, the extractable Zn further decreased but at a slower rate. This behavior could be a result of an initial rehydration of the soil, followed by ionic Zn micropore diffusion, and further complexation in soil by organic matter or the incorporation/precipitation of Zn in/onto stable oxide mineral phases.^{24,25}

In the case of B-ALG6-amended soil, there was a small increase of the Zn extractability from day 1 to day 2. This was followed by a period of slow decrease of the Zn extractability until a plateau was reached at the end of 15 days. The increase in Zn extractability in the first 2 days suggested that ZnO NPs were becoming more extractable as a result of the acid-promoted dissolution of ZnO NPs (eq 1).



At the end of 30 days of incubation after soil dosing, the extractable Zn for B-ALG6-amended soil was still lower than that for ZnCl₂-amended soil, suggesting that ZnO NPs dissolution was not complete in this period of time. It was

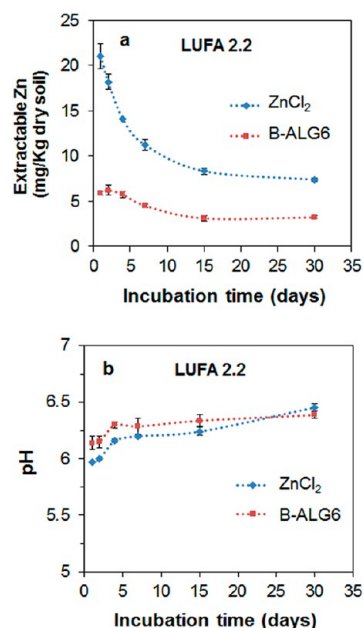


Figure 7. (a) Extractable Zn from soil LUFA 2.2 amended with ZnCl₂ and B-ALG6 as a function of time. (b) pH values of the extracts. To better visualize the trend of the results, the points in the graphics were joined with dotted lines. Error bars represent the standard deviation of the mean ($n = 3$).

also possible to observe that the pH of the extracts of B-ALG6-amended soil in the first 15 days was higher than that for soil amended with ZnCl₂ (Figure 7b). This can also be attributed to the acidic dissolution of ZnO NPs, as described by eq 1.

Similar trends were observed for LUFA 2.1 amended soils (Figure 8 a). In the soil amended with ZnCl₂, 88% of the Zn added to the soil was extractable in the first day, while in the

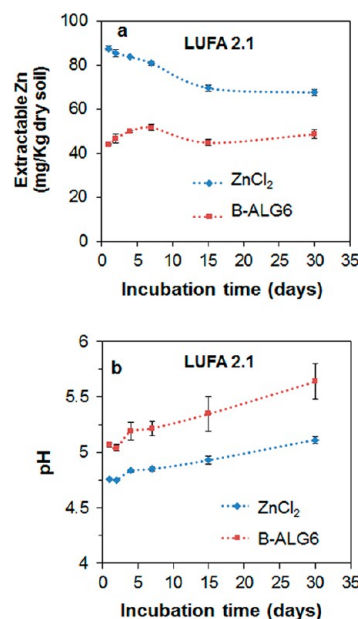


Figure 8. (a) Extractable Zn from soil LUFA 2.1 amended with ZnCl₂ and B-ALG6 as a function of time. (b) pH values of the extracts. To better visualize the trend of the results, the points in the graphics were joined with dotted lines. Error bars represent the standard deviation of the mean ($n = 3$).

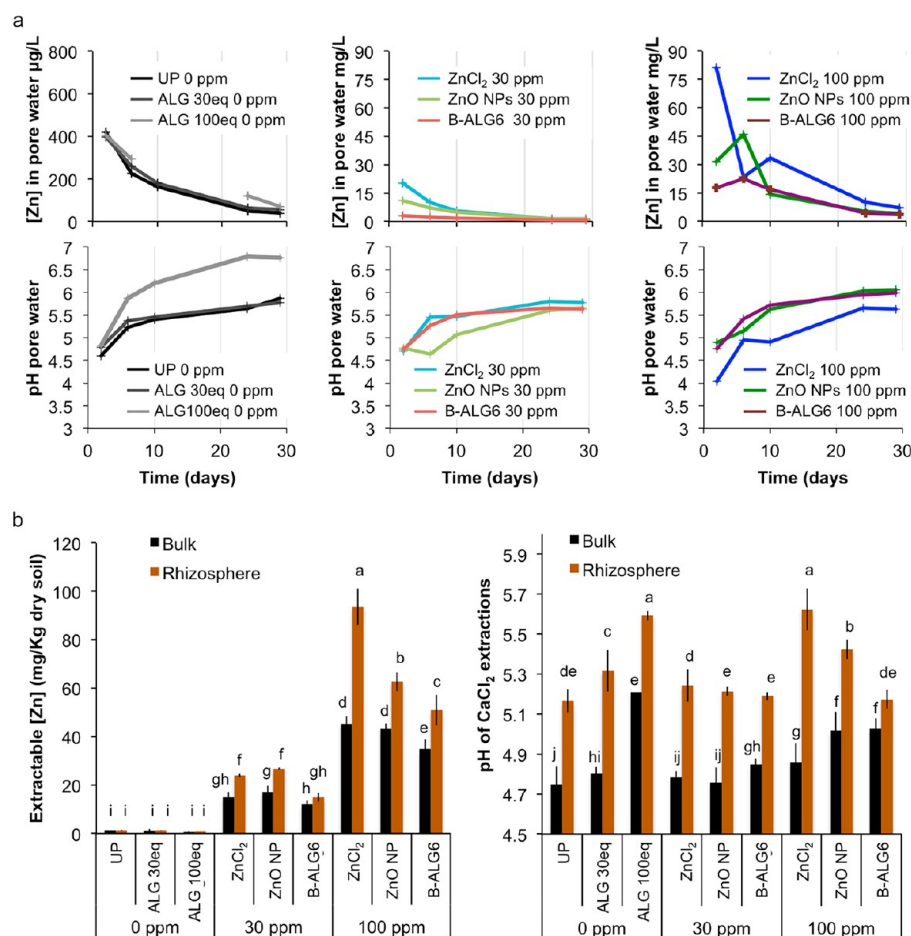


Figure 9. (a) Zn concentration in LUFA 2.1 pore water along with its pH over time of maize growth for 29 days. (b) Extractable Zn at day 29 in both the bulk and rhizosphere soils along with its pH. UP is the control. ALG 30eq and ALG 100eq are LUFA 2.1 soils dosed with ALG equivalent to those of the composite treatments. Zn was amended as ZnCl₂, ZnO NPs, and B-ALG6 at 30 or 100 mg of Zn per 1 kg of dry soil. Different letters indicate of different means (ANOVA, $p < 0.05$).

soil amended with B-ALG6, only 44% was extractable. The ZnCl₂-amended soils showed a decrease in the extractable Zn over 15 days, whereas the B-ALG6-amended soils presented a small increase in the extractable Zn in the first 7 days, and after that, the concentration of extractable Zn was almost constant. For soil LUFA 2.1 amended with B-ALG6, the period of time where an increase of the Zn extractability was registered was higher (7 days, versus 2 days in soil LUFA 2.2). This was attributed to the more acidic nature of this soil (pH 5.2), which can promote faster dissolution of ZnO NPs (eq 1).

As shown in Figures 7 and 8, the difference between the pH values of the extracts of B-ALG6- and ZnCl₂-amended soil was higher in soil LUFA 2.1 than in soil LUFA 2.2. This could be related with degradation of the polymeric matrix in this soil with the formation of products with alkaline character. Kanakaraju et al.⁵⁴ also reported an increase of the pH of a solution containing alginate beads with time, and the degradation of alginate was pointed out as one of the possible causes for the pH rise.

3.4. Controlled Release of Zn from ZnO NPs/Alginate Beads for Maize Fertilization in Acidic Soil. To investigate the potential of ZnO NPs alginate beads (B-ALG6) to be used in Zn-deficient acidic soil as a Zn source for maize, the growth of maize was assessed in LUFA 2.1 supplied with different Zn sources (ZnCl₂, ZnO NPs, or B-ALG6), at 30 or 100 mg of Zn per 1 kg of dry soil. Controls (soil amended with ultrapure

water or with a mass of alginate equivalent to that amended in the B-ALG6 treatment) were also run in parallel.

As described for the extractable Zn in soils incubated without plants, adding Zn to LUFA 2.1 increased the Zn concentration in pore water for all days in comparison to the non-Zn controls (Figure 9a and Table S5). While this increase was high at day 2, it stabilized at day 29 and it was 2–4 times higher for all of the 30 mg/kg amendments and 9–17 times higher for the 100 mg/kg amendments (Table S5). The Zn concentrations in pore water decreased from day 2 to day 29 for all conditions. While this decrease was the highest for ZnCl₂ treatments (94% and 91% decreases for the 30 and 100 ppm amendments, respectively), the B-ALG6 treatment was the one that showed the lowest decrease (72% and 79% decreases for the 30 and 100 ppm treatments, respectively), indicating that the loss of Zn by leaching in soil for that treatment was lowered (Table S5). This decrease in the Zn pore water concentrations over time seems to be driven by its pH increase (Figure 9a) because these traits significantly correlate together (Figure S15a).

The CaCl₂-extractable Zn concentrations in the bulk soil (far away from the main root systems) follow similar trends, with higher Zn amendment increasing the extractable Zn, following the trend ZnCl₂ = ZnO NPs > B-ALG6 (Figure 9b).

These differences among treatments were even clearer in the rhizosphere soil, where all of the overall extractable Zn

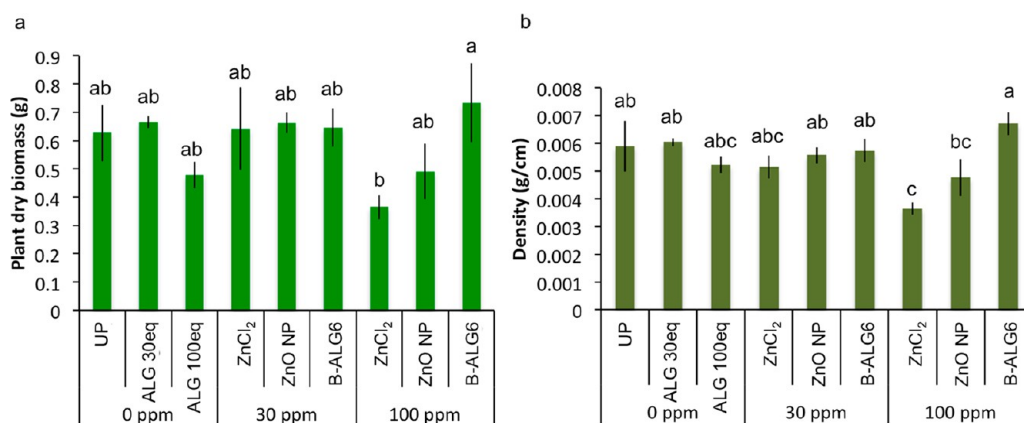


Figure 10. Maize after 29 days of growth in LUFA 2.1 amended with different treatments: (a) their biomass; (b) their density (shoot biomass normalized by the leaf length). Different letters indicate of different means (ANOVA, $p < 0.05$).

concentrations are higher than those in the bulk soil. This highlights the great influence of root exudation on the Zn bioavailability in the rhizosphere. It is noteworthy that in the rhizosphere, all of the pH values (1:10 soil/0.01 M CaCl₂) were also significantly higher than those in the bulk soil (Figure 9 b), which was surprising given the higher extractable Zn concentrations. Indeed, while extractable Zn concentrations were negatively correlated with the pH in the bulk soil, this correlation was positive in the rhizosphere soil (Figure S15b). This indicates that plants and/or their microbiome can release compounds that increase the Zn bioavailability in a non-pH-dependent manner in the rhizosphere soil, such as via the exudation of exopolysaccharides, organic matter, organic acids, and/or chelators.⁵⁵ Similar observations have been made while studying the Cu bioavailability in the rhizosphere soil of wheat.⁵⁶ This underlies the need for considering the fate of (micro)nutrients in the rhizosphere soil (fate at the root–soil interface), when studying the soil physical–chemical parameters influencing the (micro)nutrients bioavailability. Reactions in the rhizosphere soil will likely be different from those in the bulk soil, changing the predictions⁵⁷ of micronutrient bioavailability and biogeochemical cycling in the latter when plants are present.

The highest pH (>5.5) observed for ZnCl₂ and ZnO NPs 100 ppm could be a response of the plant to a (too) high bioavailable Zn concentration in its root vicinity,⁵⁵ which induces toxic impacts on the plant growth. Indeed, the dry biomass of the maize plant grown on these soils seems to validate this hypothesis. Maize growing on the soils presenting the highest extractable Zn (ZnCl₂ and ZnO NPs dosed at 100 mg/kg) produced less biomass than the control without Zn addition (Figure 10). The smaller plants and biomass observed for the alginate control could be due to an increase of the pH induced by the presence of alginate, which may reduce the bioavailability of other micronutrients. These results illustrate well how too low of a micronutrient bioavailability impairs plant growth, while too much of it induces toxic impacts.

Of all of the treatments, B-ALG6 100 ppm was the only one showing an increase in the plant biomass and density, slightly improving the plant growth. It is noteworthy that the fact that this increase was not statistically significant was likely because the experiment was stopped at 29 days, while Zn deficiency effects on the biomass should appear in the latter stages of the plant growth. This could explain why significant differences in biomass were not observed in the B-ALG6-treated soil in

comparison with the control soil, while the conditions for Zn deficiencies in LUFA 2.1 were met (Zn pore water <0.5 mg/L¹⁴ and <1 ppm of diethylenetriamine pentaacetate-extractable Zn).²³ However, some early signs of Zn deficiency (chlorosis and yellow margins) were observed on the maize leaves growing on LUFA 2.1 soil (Figure S16), which did not appear when grown on the Zn-amended soils. Signs of toxicity were observed for the ZnO NPs and ZnCl₂ treatments at 100 mg/kg concentration (Figure S16). From all of the treatments, B-ALG6 dosed at 100 mg of Zn per 1 kg of soil was the only treatment that slightly improved plant growth, while not presenting any signs of Zn deficiencies or metallic toxicity.

4. CONCLUSIONS

These results illustrate how precise the strategies for fertilization have to be to meet plant (micro)nutrient needs, while avoiding waste generation, water contamination, metal leaching in the environment, and toxic effects on soil organisms. This study demonstrates that the immobilization of ZnO NPs onto natural polymers is a way to produce materials that release Zn in soils in a controlled manner. The composites of biopolymers and ZnO NPs produced here showed lower Zn release in acidic soil than ZnCl₂ or ZnO NPs alone, with Zn release that was soil- and polymer-type dependent. Our results suggest that ZnO NPs immobilized onto alginate beads (prepared by cross-linking with CaCl₂) can be used for the controlled release of Zn in agricultural soils. ZnO NPs/alginate beads could thus meet the Zn needs of maize plant on the acidic soil LUFA 2.1. In comparison, conventional salts of ZnCl₂ have to be added in the soil in high quantities to meet the plant needs after its leaching into the soil, which can induce toxic responses at the earlier stage of plant growth and represent a large waste of materials.

This study demonstrates that fertilizers that can slowly release micronutrients in a controlled manner, such as the one presented here, can supply enough Zn for a plant to grow well while reducing Zn losses. Because of this long-term release, this kind of material could be applied once in the soil and be reused for several crop cycles.

Further studies of Zn release from these types of materials should be performed for longer times to better understand the fate of these materials in the bulk soil in comparison to the rhizosphere soil, to ensure the safety of materials for use in agrifood production. The scalability and cost are also important parameters that must be considered when analyzing

the viability of such new fertilizers from a commercial point of view.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsanm.9b01492>.

SEM and TEM images and particle size distribution histogram for commercial ZnO NPs, XRD pattern for ZnO NPs and ZnO composites, UV-vis spectra of ZnO NPs and ZnO composites, EDS spectrum of MCC-ZnO₁, TEM image of MCC-ZnO₂, SEM images of MCC-ZnO₂ ALG and MCC-ZnO₃ ALG, FTIR-ATR spectra of G-CH-ZnO₅, photographs of alginate beads and SEM images of the interior of the beads, pH values of suspensions of all materials, extractable Zn from soils DE and VV amended with B-ALG6 or ZnCl₂ (30 day incubation test), photographs showing Zn deficiencies in maize growth, correlations between the Zn concentration in pore water or bulk soil or rhizosphere soil and the pH, and decrease of the Zn concentration in soil pore water between day 2 and day 29 for different treatments (PDF)

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Notes

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